

COMMUNICATION PROTOCOL

UAS MINI PROJECT

Credit Score Mock API

Enterprise Credit Scoring REST API dengan Observability Dashboard — mensimulasikan proses credit scoring menggunakan REST API dengan fitur audit log, Request ID, Correlation ID, response time, dan status code.

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Latar Belakang

Problem Statement

Lembaga keuangan membutuhkan sistem yang mampu menilai risiko calon peminjam secara cepat, konsisten, dan terdokumentasi — menerima data pengajuan, melakukan penilaian, menghasilkan keputusan kredit, serta menyediakan mekanisme monitoring terhadap seluruh aktivitas API.

REST API

Membangun REST API sederhana menggunakan TanStack Start

HTTP + JSON

Implementasi komunikasi HTTP berbasis JSON

Postman

Pengujian endpoint menggunakan Postman

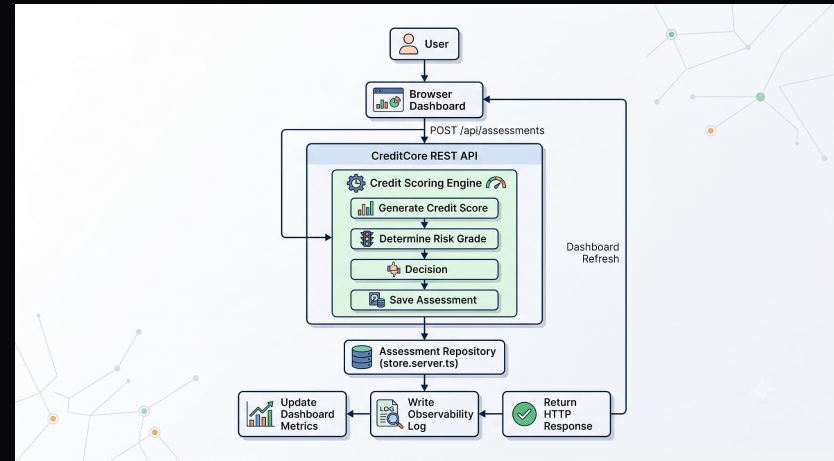
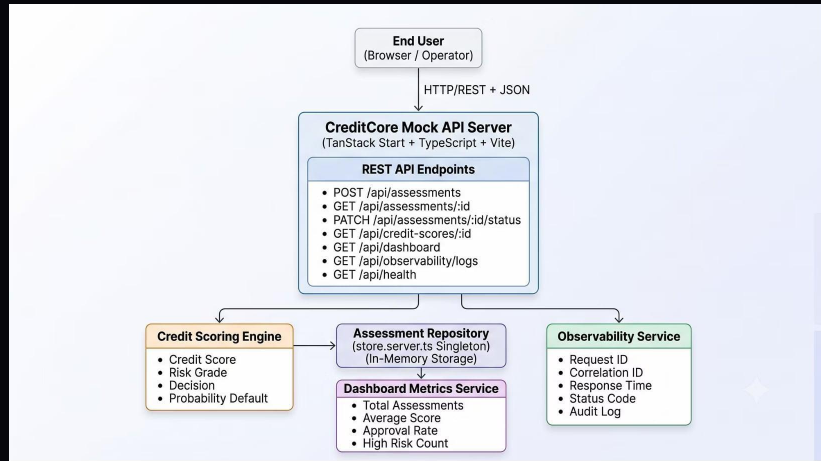
Observability

Monitoring terhadap seluruh request API

Use Case: Model Scoring Mock API

Arsitektur Sistem

Sistem terdiri dari End User, REST API Server, Credit Scoring Engine, Assessment Repository, Dashboard Metrics Service, dan Observability Service. Client mengirim request via HTTP/REST, server memproses melalui Credit Scoring Engine, menyimpan hasil ke repository, dan dashboard menampilkan statistik.



Endpoint Design

Business & Infrastructure API

Method	Endpoint	Fungsi
POST	/api/assessments	Membuat assessment baru
GET	/api/assessments/:id	Detail assessment berdasarkan id
PATCH	/api/assessments/:id/status	Update status assessment
GET	/api/credit-scores/:id	Melihat credit score
GET	/api/health	Memeriksa status server
GET	/api/dashboard	Menampilkan statistik dashboard
GET	/api/observability/logs	Menampilkan audit log

Postman Collection juga menyediakan endpoint simulasi (GET/PUT /api/simulation) untuk menguji skenario error seperti HTTP 500 tanpa mengubah kode aplikasi.

Success Scenario

Response Berhasil

POST /api/assessments — 201 Created

The screenshot shows the Credicore Mock API interface. The left sidebar lists collections: CreditCore - Mock API, Infrastructure, Health, Dashboard Metrics, Observability - Logs, Simulation - Get, Simulation - Trigger 500, Simulation - Reset, Business API, Create Assessment, List Assessments, Get Assessment by ID, and Get Credit Score. The main panel displays the details of a POST request to `/api/assessments`. The request body is a JSON object with fields like `assessmentId`, `status`, `createdAt`, `updatedAt`, `requestId`, `requestInfo`, `latencyMs`, `year`, `method`, and `source`. The response status is 201 Created, and the response body is a JSON object with fields like `success`, `data`, `assessmentId`, `status`, `createdAt`, `updatedAt`, `requestId`, `requestInfo`, `latencyMs`, `year`, `method`, and `source`.

GET /api/dashboard — 200 OK

The screenshot shows the Credicore Mock API interface. The left sidebar lists collections: CreditCore - Mock API, Infrastructure, Health, Dashboard Metrics, Observability - Logs, Simulation - Get, Simulation - Trigger 500, Simulation - Reset, Business API, Create Assessment, List Assessments, Get Assessment by ID, and Get Credit Score. The main panel displays the details of a GET request to `/api/dashboard`. The response status is 200 OK, and the response body is a JSON object with fields like `totalAssessments`, `averageCreditScore`, `approvalRate`, `approvalCount`, `averageMonthlyIncome`, `lightAssessments`, `totalRequests`, `completedCases`, `secretAssessments`, `accountInfo`, `secretName`, `requestAccount`, `monthlyIncome`, `employmentLengthMonths`, and `purposeOfLoan`.

Failure Scenario

Response Kegagalan

Get /api/assessments — 404 Not Found

The screenshot shows the Burp Suite interface with the following details:

- URL:** GET `https://[redacted]/api/assessments`
- Method:** GET
- Response:** 404 Not Found - 5 ms - 908 B
- Body:** A JSON response indicating a 404 error. The response structure is as follows:

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "RESOURCE_NOT_FOUND",
    "message": "Unknown API endpoint: GET /api/assessments"
  },
  "meta": {
    "requestId": "6206dee-0003-4206-e890-0170ca0001a",
    "correlationId": "4acc6a3c-7746-42e7-a44b-23109d6a3370",
    "timestamp": "2026-07-12T07:19:18.566Z",
    "latency": 5,
    "path": "/api/assessments",
    "method": "GET",
    "source": "Postman"
  }
}
```

Post /api/assessments — 400 Bad Request

The screenshot shows the Burp Suite interface with the following details:

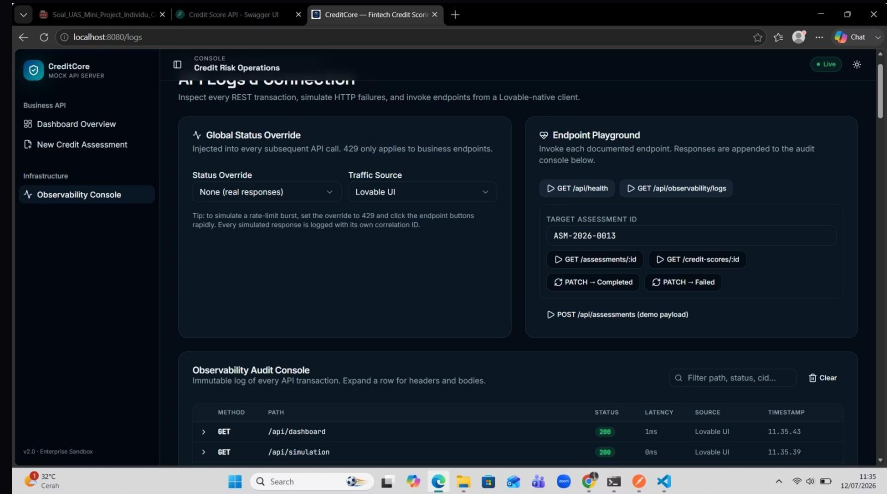
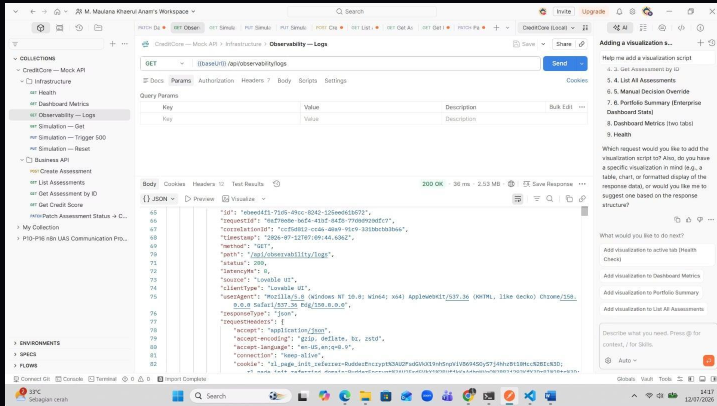
- URL:** POST `https://[redacted]/api/assessments`
- Method:** POST
- Response:** 400 Bad Request - 17 ms - 908 B
- Body:** A JSON response indicating a 400 error. The response structure is as follows:

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "DATA_CONTRACT_VIOLATION",
    "message": "Payload must be a JSON object."
  },
  "meta": {
    "requestId": "420601ac-6d3d-4dfe-aee7-47603434e01a",
    "correlationId": "603030a3-690c-484d-b06f-460093006339",
    "timestamp": "2026-07-12T07:19:13.496Z",
    "latency": 6,
    "path": "/api/assessments",
    "method": "POST",
    "source": "Postman"
  }
}
```

Observability Evidence

Audit Log & Monitoring

Sistem menyediakan observability melalui endpoint `GET /api/observability/logs`. Setiap request mencatat informasi lengkap untuk monitoring dan debugging real-time.



Wireshark Capture

Analisis Paket Jaringan

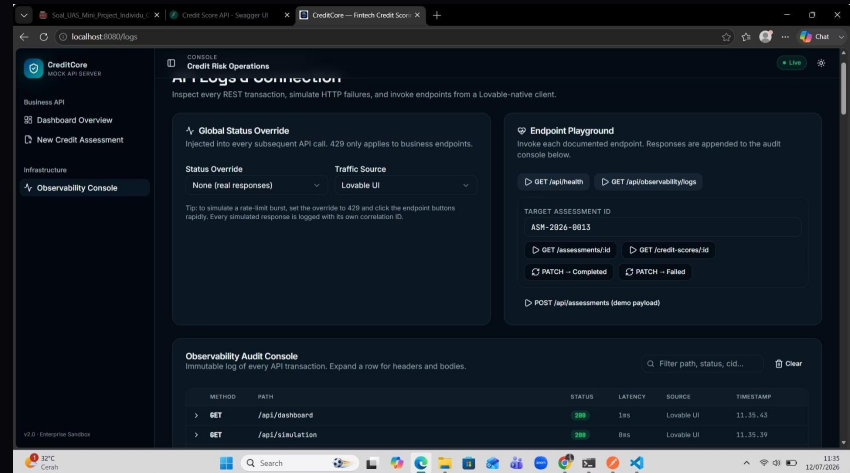
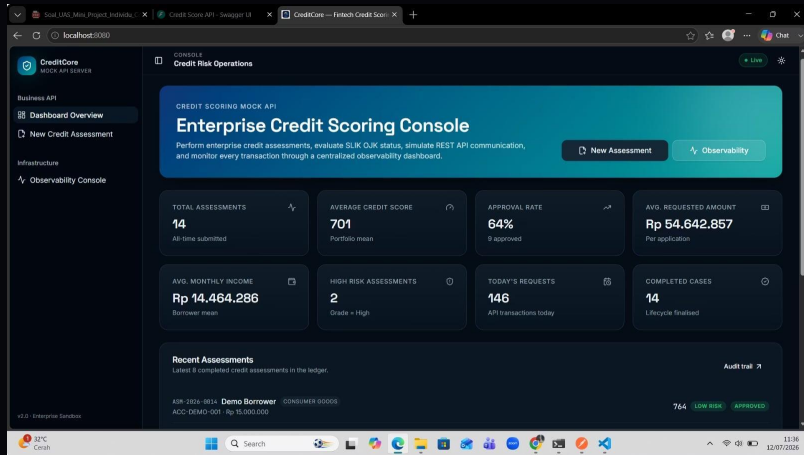
Port 8080 digunakan oleh server CreditCore Mock API. Karena server dijalankan pada *localhost*, seluruh paket dikirim melalui *loopback interface* dengan source dan destination sama-sama : 1.

Wireshark capture of loopback traffic on port 8080. The packet list shows a series of TCP and HTTP packets between 127.0.0.1. The selected packet (No. 11) is an HTTP GET request to /api/observability/logs. The packet details pane shows the structure of the HTTP request, including the status bar, request line, headers, and body. The packet bytes pane shows the raw data in hexadecimal and ASCII.

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Demo App

Pada demo ini ditampilkan aplikasi Credit Score Hub yang berfungsi sebagai mock credit scoring platform. Dashboard mengambil data langsung dari backend melalui REST API.



Testing Result & Conclusion

Hasil Pengujian & Refleksi



Pengujian Lulus

Create, Get, Dashboard, Credit Score, Observability, Health, Error Testing



Status Code

200 OK, 201 Created, 400 Bad Request, 404 Not Found



Reliability

Response sesuai kontrak API dengan audit log lengkap

Limitation

- In-Memory Storage, data hilang saat server berhenti
- Belum menggunakan database PostgreSQL/MySQL
- Belum menerapkan autentikasi & otorisasi
- Belum menggunakan HTTPS

Improvement

- Integrasi database PostgreSQL
- Autentikasi JWT
- Docker untuk deployment
- Workflow otomatisasi n8n
- Rate limiting & monitoring komprehensif

GitHub: github.com/khaerulanamm/creditscorehub • **Video Demo:** youtube.com/xxxxxxx